



IRA A. FULTON COLLEGE OF ENGINEERING

LABORATORY & SHOP SAFETY PLAN

INCLUDES CHEMICAL HYGIENE PLAN



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1.0 OVERVIEW

This plan outlines the systematic process laboratories and shops are to incorporate to ensure the safety of individuals working in their areas. *Note: OSHA's required "Chemical Hygiene Plan" has been incorporated into this document.*

Individuals who manage a laboratory or shop help keep those they supervise safe by overseeing the completion of:

- Job Hazard Analysis
- Written Standard Operating Procedures (SOP's)
- Proper training
- Safety reviews

Laboratories and shops do not need to develop written Standard Operating Procedures (SOP's) for simple tasks where consumer products are used for the purpose intended by the manufacturer so long as they adhere to manufacturer use instructions. Examples include, using Windex to clean the windshield of a vehicle, filling the oil reservoir of a vehicle with oil, and using Fast Orange to clean grease from a part in a parts washer.

2.0 APPLICABLE STANDARDS

1. OSHA General Industry Standards (29 CFR 1910)
2. NFPA 70E
3. ASME Boiler and Pressure Vessel Code

3.0 PURPOSE

The purpose of this plan is to standardize how safety is managed in laboratories and shops, with the goal being to prevent fatalities and significant injuries.

4.0 SCOPE

The requirements found in this plan apply to all laboratories and shops within the College of Engineering.

Note: when this plan refers to hazardous chemicals it is referring to chemicals referred to as hazardous by OSHA (see 29 CFR 1910.1200). The definition is provided in the Chemical Safety and Handling Section.

5.0 OVERVIEW OF SYSTEMATIC PROCESS

All work within laboratories and shops must be assessed for hazards prior to engaging in the work and then reassessed as necessary. Once hazards have been identified and evaluated, for a particular job or task, appropriate hazard controls are to be selected and implemented, and individuals trained so they know how to perform their work safely. This process is outlined as follows:



When the work being performed changes then the Job Hazard Analysis (JHA) and Standard Operating Procedure (SOP) associated with the work must be reassessed and updated as necessary.

More detail regarding each of the steps is provided in subsequent sections of this document. In addition to incorporating the process outlined above, laboratories and shops must also ensure other applicable University program and policy requirements are satisfied and existing safety regulations are complied with.

6.0 JOB HAZARD ANALYSIS (JHA)

Before performing a Job Hazard Analysis make sure you have addressed any **serious hazards you already know about**. Don't wait to address such hazards.

-Steps Needed to Prepare for a Job Hazard Analysis-

- A. Create a list of jobs** for the work being performed in your lab or shop. Then prioritize your list based upon perceived hazard. Address the most hazardous job before moving to the next until all jobs have had a Job Hazard Analysis performed for it.
- B. Form a Review Team to help perform the Job Hazard Analysis.** This team will help identify hazards, evaluate hazards, and help identify proper mitigation and control measures. At minimum you should involve a person who knows how to perform the

work, a safety professional, a subject matter expert, and an individual who can commit financial resources for hazard controls. *Note: an individual who can commit financial resources could also fill any of the other roles.*

The following steps outline how to perform a JHA. Steps 1 and 2 can be performed by a single person but the other steps need to involve the Review Team:

1. **Provide the job being analyzed with an appropriate title and description.** The title and description for each job should adequately describe the work that is occurring.
2. **Divide the job into tasks.** It's up to you how many tasks you want to identify. You're dividing the job into tasks to make analysis more manageable by addressing smaller pieces of the whole.
3. **Identify all hazards associated with each task.** You'll need to consider hazards that may occur under normal and abnormal operating conditions. Some hazards are obvious whereas others are not. A hazard checklist can be helpful therefore one has been provided in Appendix A of this plan.

Be sure to describe each hazard well enough to relay the type of hazard being encountered. *Examples include biological, chemical, noise, radiation, moving parts, etc.*

4. **Evaluate the Hazards.** Each hazard needs to be evaluated for likelihood of occurrence and severity of outcome. There are many Hazard Evaluation methods available, including, but not limited to:

Qualitative Methods	Quantitative Methods
Preliminary Hazard Analysis (PreHA)	Layer of Protection Analysis (LOPA)
Hazard & Operability (HAZOP) Analysis	Fault Tree Analysis (FTA)
What-If Analysis	Event Tree Analysis (ETA)
2 GuideWord Analysis	Human Reliability Analysis (HRA) event tree
Failure Modes & Effects Analysis (FMEA)	
Checklist (traditional)	

The Qualitative Methods find scenarios, whereas the Quantitative Methods can help you decide how best to lower risk. Appendix C provides more information about these evaluation methods.

The correct Hazard Evaluation method is the one that is matched to the complexity and hazards associated with the process. The Review Team is to select the most appropriate method to use. The objective is to prevent fatalities and injuries and protect University property.

The likelihood and severity of a hazard's occurrence will need to be combined to determine a risk rating. The risk ratings for hazards will help the Review Team determine if there are any hazards that don't need to be addressed and prioritize which hazards need to be addressed first. A risk matrix like the following can be used:

Safety & Health Risk Assessment Matrix

Increasing Probability of Adverse Outcome ⇨

	A	B	C	D	E
Consequence	Unlikely	Seldom	Occassional	Likely	Frequent
No Injury	I	I	I	I	I
Slight Injury	I	I	I	I	I
Moderate Injury	I	I	I	II	II
Serious Injury	I	II	II	II	III
1 -3 deaths	II	II	III	III	III
Many Deaths	III	III	III	III	III

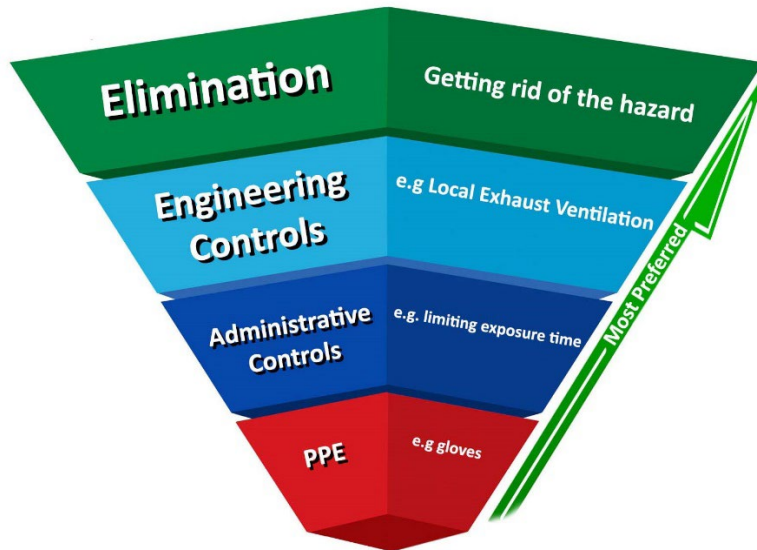
Increased Risk ⇨



Orange and red risk means a higher degree of assessment is needed
(e.g. What-if Analysis, HAZOP, Fault Tree Analysis, ...)

It's possible that hazards with a risk rating of I in the green may not need to be addressed. Other hazards need to be addressed to one degree or another, with category III being the highest priority.

- Determine how the hazards will be eliminated or controlled.** The goal is to identify proper controls, safeguards, and other layers of protection needed to prevent fatalities and injuries while protecting University property. The hierarchy of preferred hazard control measures needs to be followed when deciding upon what controls to implement.
Note: the need to use more than one control measure at a time is not uncommon.



The most effective and desirable control measures minimize the potential for human error. This is why the use of personal protective equipment is the least desirable. It relies heavily upon people putting it on, wearing it properly, and maintaining it properly. In contrast a good engineering control can do its job by simply being present...like a guardrail system used to protect people from falling. It is less important that you understand the definition of each type of control measure, and more important to understand there is a hierarchy to follow and personal protective equipment is not the first choice.

Hazard mitigation and control measures must at very least satisfy existing regulatory standards. Where these are lacking mitigation and control measures need to be benchmarked against existing U.S. industry practice.

6. **Adequately document your Job Hazard Analysis (JHA).**
7. **Implement the hazard control measures you've identified.**

-Periodic Review- Make sure to keep your Job Hazard Analysis up to date. This means if the work changes then a review is needed. A periodic review is also needed on a recurring basis to ensure accuracy and thoroughness. At minimum, review JHA's every other year.

-Suggestion- maintain copies of completed Job Hazard Analysis (JHA) for common jobs electronically on your department website where they can be reviewed by you, peers, other subject matter experts, and safety professionals.

7.0 STANDARD OPERATING PROCEDURES

Individuals need to know how to perform their work safely. To do so they will need to complete adequate training. Completed Job Hazard Analysis (JHA), Safety Data Sheets (SDS), and Standard

Operating Procedures (SOPs) provide the basis for laboratory specific training. Standard Operating Procedures must be created by laboratories and shops with the following exception:

Laboratories and shops do not need to develop written Standard Operating Procedures (SOP's) for simple tasks where consumer products are used for the purpose intended by the manufacturer so long as they adhere to manufacturer use instructions. Examples include, using Windex to clean the windshield of a vehicle, filling the oil reservoir of a vehicle with oil, and using Fast Orange to clean grease from a part in a parts washer.

Standard Operating Procedures can be in the format preferred by the laboratory or shop. Appendix B of this plan provides a possible SOP format. Manufacturer instructions for how to use a machine or piece of equipment can be adopted as a Standard Operating Procedure when appropriate.

Standard Operating Procedures (SOP's) need to include a section regarding what to do if an emergency occurs. For instance, a lab may need to shut a piece of equipment down during the event of a building fire to ensure a greater disaster does not occur.

-Periodic Review- As with JHA, make sure to keep your Standard Operating Procedures (SOP) up to date. This means if the work changes, then a review of the relevant SOP is needed. A periodic review is also needed on a recurring basis to ensure accuracy and thoroughness. At minimum, review existing SOP's for ongoing work every other year.

8.0 TRAINING

All individuals must complete training sufficiently to ensure they know how to perform their work safely. Such training is to be based upon completed Job Hazard Analysis (JHA), Standard Operating Procedures (SOP's), and applicable Safety Data Sheets (SDS). *Note: or manufacturer instructions when appropriate.*

Note: The College of Engineering, Risk Management, and other BYU organizations have created a library of safety training courses that are provided on BYU's website. These courses can be used to ease the training burden placed on supervisors.

9.0 CHEMICAL SAFETY AND HANDLING

"Hazardous chemical" means a chemical for which there is statistically significant evidence based on at least one study conducted in accordance with established scientific principles that acute or chronic health effects may occur in exposed employees. The term "health hazard" includes chemicals which are carcinogens, toxic or highly toxic agents, reproductive toxins, irritants, corrosives, sensitizers, hepatotoxins, nephrotoxins, neurotoxins, agents which act on the hematopoietic systems, and agents which damage the lungs, skin, eyes, or mucous membranes." 29 CFR 1910.1450.

SAFETY DATA SHEETS (SDS)

Current copies of Safety Data Sheets, with an accompanying inventory, are to be maintained by each laboratory. This allows workers to have ready access to the information. *Note: the inventory and associated SDS can be kept electronically on a USB thumb drive.*

Please review the Safety Data Sheet (SDS) for each chemical or chemical mixture you use. Doing so will help you understand associated hazards.

EXPOSURE TO CHEMICALS

Individuals should minimize their exposure to chemical substances. The measures needed to minimize exposure are to be identified in completed Job Hazard Analysis and documented in Standard Operating Procedures.

If you are concerned that you may be exposed to too much of a hazardous chemical, please contact an Occupational Health and Safety Professional so an evaluation can be performed:

College of Engineering: 801-422-6589

Risk Management: 801-422-2943

CHEMICAL STORAGE

Hazardous chemicals must be stored according to compatibility, and in accordance with the directions provided by the manufacturer (see applicable Safety Data Sheets and product labels for directions). Also, follow these guidelines:

- Flammable chemicals are to be stored in Flammable Storage Cabinets unless permission by the College is otherwise granted;
- Do not store chemicals above shoulder level; and
- When hazardous chemicals are stored on shelves. The shelving is to be designed to prevent chemicals from falling off during an earthquake. The shelving must be secured to the wall of the lab.

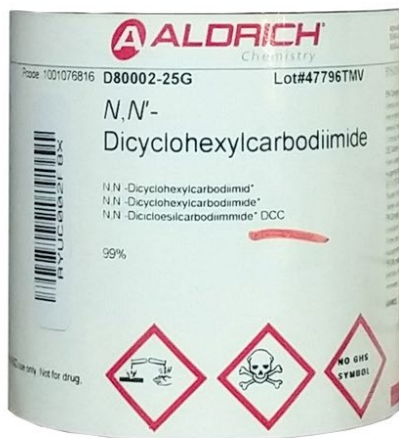
If you have questions regarding proper chemical storage, please reach out to an Occupational Health and Safety Professional.

QUANTITY OF CHEMICALS ON HAND

Minimize waste and expense by only buying and storing what you will use for the work you are performing. Order more frequently rather than buying larger containers. This practice will help reduce the risks involved with storing chemicals and minimize the amount that needs to be provided to Chemicals Management as Unwanted Lab Material when you are finished with your work. Disposal costs almost always outweigh the upfront savings associated with buying a larger container and receiving a small discount.

CHEMICAL CONTAINER LABELING

All containers purchased from a supplier, manufacturer, or online store must bear a legible primary chemical container label. Example:



Primary Container Label

(Applied by Manufacturer)

Other containers are considered secondary containers, and they must be labeled to convey what is in them and what hazards are presented by the contents. Secondary chemical containers are applied by the laboratory or shop. Example:

Chemical Name/Amount/Concentration:	
GHS pictograms (Please circle):	
Preparer's Name: _____	Date: _____

Secondary chemical containers must bear the product name. The name of the chemical substance, hazard warnings, and hazard pictograms can be found on Safety Data Sheet(s). *Note: If the secondary containers are too small to be labeled, then the container holding the small secondary container can be labeled...it must be readily apparent to emergency responders what is in a container.*

Risk Management has created an online resource to help individuals create secondary container labels. Please see Risk Management's website: <https://risk.byu.edu/>

UNWANTED LABORATORY MATERIALS

What are "Unwanted Laboratory Materials"? They are "reactive acutely hazardous unwanted materials and materials that may eventually be determined not to be a solid or a hazardous waste." (40 CFR 262.200) - basically anything generated by a laboratory or shop that someone might be tempted to throw into the trash or down the drain that isn't common trash. Provide all Unwanted Laboratory Materials to Chemicals Management.

If you have questions regarding collection and getting rid of Unwanted Laboratory Materials, please contact Chemicals Management (801-422-6382).

CHEMICAL SPILLS AND SPILL KITS

Chemical spills/releases are either cleaned up by the laboratory or shop (if safe to do so), by Risk Management, by the Fire Department, or by a contractor. If a laboratory or shop is not prepared or shouldn't clean up a spill then they need to evacuate the area and contact Risk Management (801-422-4468) or the University Police (801-422-2222). Proper response needs to be outlined in applicable Standard Operating Procedure(s).

Laboratories and shops are to maintain a spill kit adequate for the size of spill(s) they can safely clean up.

OSHA asks that employees review the OSHA Laboratory Standard which can be found on the OSHA website:

http://www.osha.gov/pls/oshaweb/owadisp.show_document?p_table=STANDARDS&p_id=10106

10.0 ADDITIONAL REQUIREMENTS AND GUIDANCE

ELECTRICAL SAFETY

Electrical hazards are present throughout the University. Such hazards could lead to electrical shock, electrocution, or arc flash. Therefore wiring, equipment, conductors, and circuit parts (energized to 50-volts or higher) must be designed and constructed in accordance with OSHA and NFPA requirements; and used according to manufacturer recommendations.

Electrical equipment must be in safe working condition, and must be "Approved" and "Accepted" per OSHA <https://www.osha.gov/laws-regs/regulations/standardnumber/1910/1910.399>

Individuals designing, constructing, servicing and otherwise working on wiring, equipment, conductors, and circuit parts must be qualified to do so. "Qualified" per NFPA 70E. The work must be done in compliance with the University Electrical Safety Program, OSHA Subpart S requirements, and NFPA 70E.

Work on or near exposed energized parts (50-volts and higher) is not allowed unless there is no other option and the requirements mentioned in the previous paragraph are satisfied. Circuits and electrical equipment must be placed into an electrically safe working condition, which means completely de-energized and locked out.

Exception - Work on or near exposed energized parts (50 volts and higher) is sometimes necessary (e.g. diagnostics and work on batteries). When this is the case the work is to be performed in accordance with the University Electrical Safety Program, OSHA Subpart S requirements, and NFPA 70E.

IONIZING RADIATION

If you are planning to use ionizing radiation, please contact the University Radiation Safety Officer (RSO). Sources of ionizing radiation can include equipment, machines (e.g. x-ray

machine), and radioisotopes (e.g. carbon-14). Only Principal Investigators (PIs) approved by the University Radiation Safety Officer (RSO) may order radioactive materials.

Time, distance, and shielding will be used to reduce exposure to ionizing radiation so exposures are as low as reasonably achievable, and all work will be done in accordance with the University Radiation Safety Program and State and Federal regulations. Please contact the University Radiation Safety Officer for further guidance (801-422-6879).

LASER SAFETY

The use of any laser needs to occur in a controlled manner to protect all individuals who could potentially be exposed to the beam.

Class 1 and Class 2 laser products are to be used according to the guidance provided by the manufacturer. Laser pointers are never to be directed into the eye of any individual.

Higher powered lasers (Class 3 and 4) must be used in an area that has been set up properly. This controlled area is to satisfy the most recent version of ANSI Z136.1 requirements. The best approach is to enclose the beam so it can't be directed into the eye of any individual. Warning signs and other measures will be needed. Please reference the University Laser Safety Program and contact the University Laser Safety Officer (801-422-6879) for further guidance.

All lasers must bear an FDA label indicating the power and class. Lasers that don't bear an FDA label are not legal in the U.S.

BIOLOGICAL SAFETY

The University has a Biological Safety Program, which can be found on Risk Managements website. All work involving the use of biological safety level 2 or 3 agents or work involving recombinant DNA molecules must be performed according to the requirements found in the Biological Safety Program.

There is also an Institutional Biosafety Committee (IBC). If you are seeking to perform any research or classroom work involving the use of biological safety level 2 or 3 agents or work with recombinant DNA then the work will need to be approved by the committee beforehand. The IBC will ask for certain measures to be implemented prior to granting approval for the work to proceed.

Please contact Risk Managements Biological Safety Officer (801-422-2789) for further guidance.

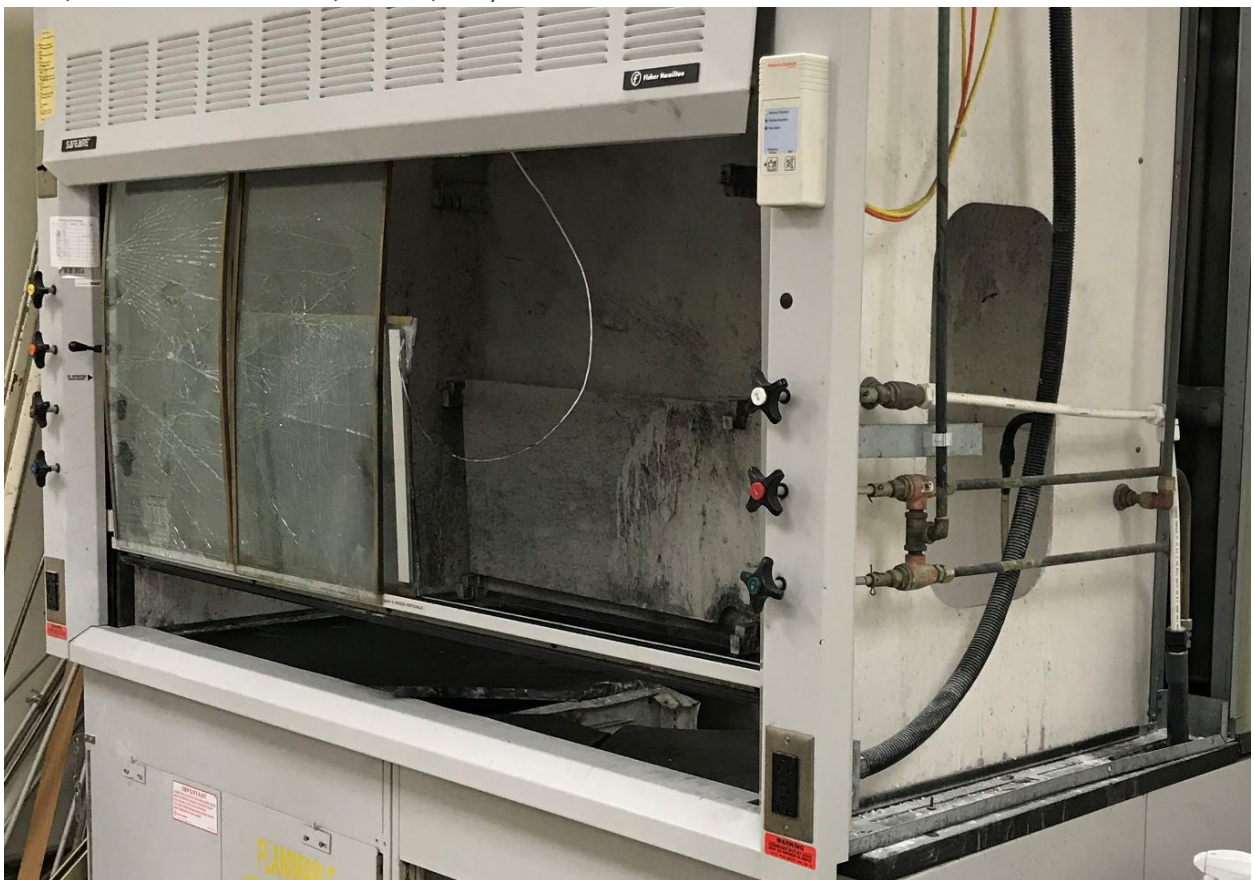
PRESSURE VESSELS

The State of Utah defines a pressure vessel as "a vessel in which the pressure is obtained from an external source or by the application of heat from an indirect source, or from a direct source

other than those boilers as previously defined.” (Boiler and Pressure Vessel Compliance Manual for the State of Utah). *Examples include reactors, glassware, autoclaves, vacuum chambers, and custom designed vessels.*

The failure of a pressure vessel can lead to serious injuries, loss of life, and significant damage to property. There are plenty of examples of catastrophic events related to the improper design and/or misuse of pressure vessels nationwide.

One instance that occurred here at BYU involved the rupture of a small pressure vessel. The user overfilled the pressure vessel, which led to over-pressurizing it as it was heated. The removable bottom plate of the reactor was violently pushed out of the bottom of the vessel, through the lab hood counter, and ejected across the room (see image below). Fortunately, no one was injured, but it should be considered a near miss, and there was significant property damage (lab hood, ventilation duct valve, reactor, etc.).



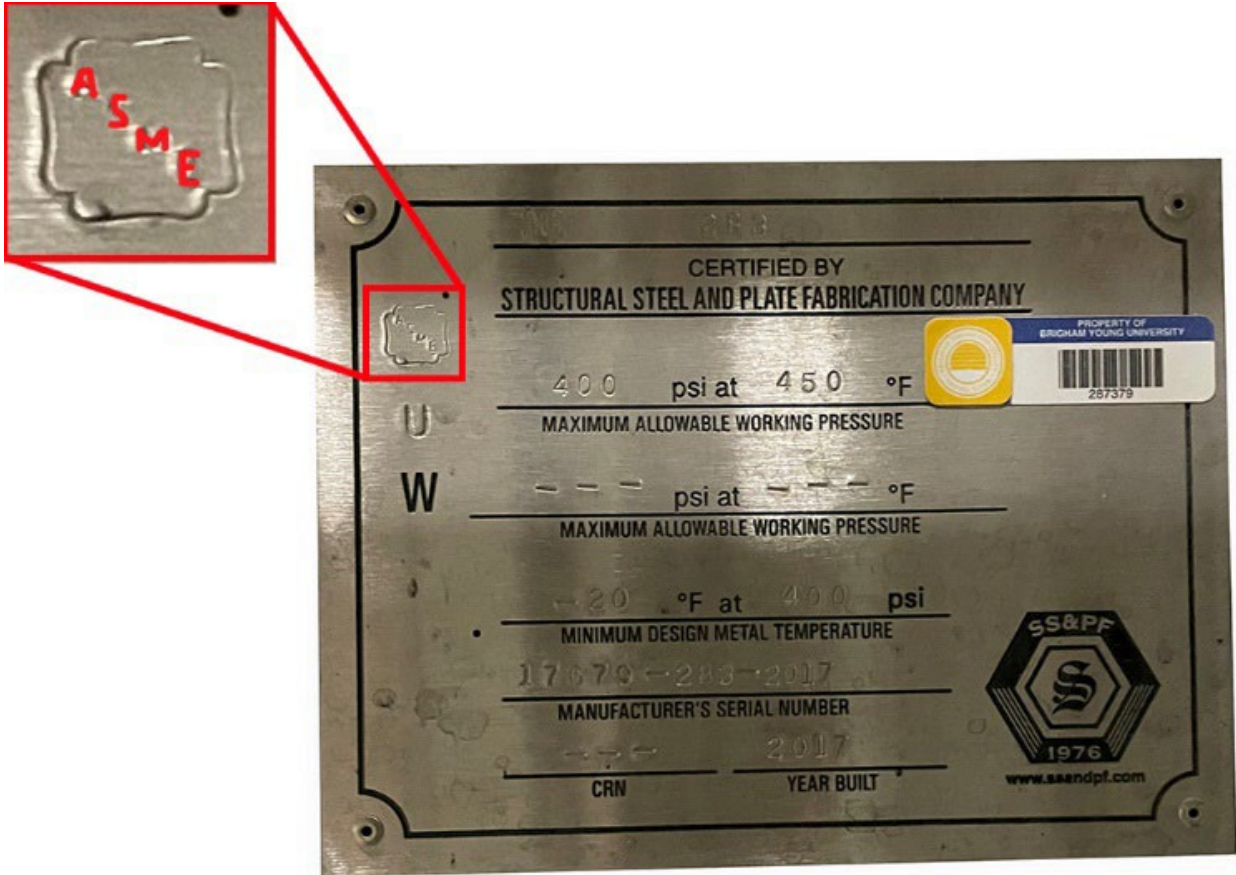


-Above- The reactor involved in the incident. Notice the warped metal at the base of the reactor.

-WHAT YOU NEED TO KNOW-

The **State of Utah Boiler and Pressure Vessel Compliance Manual** requires pressure vessels having an internal diameter of 6-inches and greater to be **inspected** and **certified** by a State Inspector (or their representative) before first use and regularly thereafter. The frequency of the regular inspections is identified by the State. If inspection and certification has not occurred, or is not occurring, then the pressure vessel can't legally be used. *Note: FM Global inspects pressure vessels on behalf of the State here at BYU. Also, compressed gas cylinders and tanks travelling on public roads are regulated by the U.S. Department of Transportation and are not subject to the provisions of the Boiler and Pressure Vessel Compliance Manual.*

Pressure vessels that have been built to satisfy ASME code will bear a nameplate indicating such (like that pictured below). Documentation that the State will look for is also provided to a buyer at the time of purchase. Maintain the documentation and don't remove the nameplate! If ASME information is found to be lacking, then the inspector will not inspect the vessel and it cannot be used legally in the State.



What about pressure vessels that have an internal diameter less than 6-inches and don't require state inspection? They must be designed and constructed to satisfy a good safe standard of construction. What options are there? The ASME Boiler and Pressure Vessel standard (BPVC Section VIII), British Pressure Vessel Standard PD 5500, BS EN 13445 European Standard, AS 1210 Australian Standard, or ISO 16528. Since we are in the U.S. it is highly recommended that you use the ASME standard when it can be used. *Warning: using a pressure vessel that is not designed to satisfy a safe standard of construction endangers you and those you work with and opens you up to liability for damage to people and property.*

If the pressure vessel is made of glass (i.e. borosilicate glass) then make sure it has been certified by a Nationally Recognized Testing Laboratory (like UL or CSA) for your application, or it satisfies AS 1210 Australian Standard for pressure vessels or the BS EN 1595 European Standard. If using a glass pressure vessel guarding and shielding of the vessel is needed.

Documentation pertaining to the standard satisfied must be maintained by the user.

PRESSURE VESSELS MUST BE OPERATED WITHIN SPECIFIED USE PARAMETERS. The manufacturer specifies use parameters and includes them in the documentation they provide. These parameters include but are not limited to maximum allowable working pressure and temperature. Compatibility also needs to be considered.

As a pressure vessel is used its strength lessens. Over time the vessel will need to be disposed of. This occurs when any of the specifications are no longer valid due to fatigue, corrosion, wear, and/or other damage. Establish a routine inspection and maintenance schedule to ensure the pressure vessel is in safe operating condition before each use. This includes monitoring corrosion. **NEVER USE A DAMAGED PRESSURE VESSEL OR ONE THAT NEEDS INSPECTION BUT HASN'T BEEN INSPECTED.**

-ADDITIONAL SAFETY MEASURES-

- Train and manage users. To ensure safety they **MUST** use the pressure vessel within design limitations.
- Create a Standard Operating Procedure (SOP) so safe operating steps are documented, can be referenced, and can be used to facilitate training. Limit and carefully control the application of heat.
- As prudent, use shielding and guards to protect users. This could be in the form of metal or shatter-proof glass or plastic screen for protection.
- Ventilate the lid and pressure relief device if release of pressure vessel contents could lead to harm. If needed, the ventilation will need to be designed properly. Please consult Physical Facilities.
- Wear appropriate personal protective equipment when working near the pressure vessel (e.g. safety glasses, face shield, ...).

If inspected by the State, the State will give the pressure vessel a U-number like that depicted on the autoclave pictured below:



If you have questions, please reach out to an Occupational Health & Safety Specialist.

Note: Compressed gas cylinders and tanks travelling on public roads are regulated by the U.S. Department of Transportation and are not subject to the State inspections outlined above.

THE RIGHT FOR MEDICAL CONSULTATION

All employees have the right to receive medical attention:

1. If after being exposed to a hazardous chemical in the laboratory and individual develops signs or symptoms associated with exposure to that hazardous chemical;
2. When exposure monitoring results for an OSHA regulated substance requires medical surveillance; and
3. Whenever an event takes place in the work area such as a spill, leak, explosion, or other occurrence resulting in the likelihood of hazardous exposure. In this case, medical consultation can be sought for the purpose of determining the need for a medical examination, which will be provided if needed.

All medical examinations and consultations shall be performed by or under the direct supervision of a licensed physician and shall be provided without cost to the employee, without loss of pay and at reasonable times and locations. Follow-up exams and/or treatment prescribed by the physician(s) will also be provided to the employee.

11.0 RESPONSIBILITIES

LABORATORY AND SHOP SUPERVISORS

- Ensure this Laboratory and Shop Safety Plan is followed and complied with.
- Ensure the work being performed is done in compliance with local and federal laws and regulations.
- Supervise individuals they are responsible for.
- Investigate near misses and/or accidents that occur in the laboratory or shop, and revise work practices and hazard control measures as necessary.

LABORATORY & SHOP WORKERS (Employees and Students)

- Complete the training required by your supervisor.
- Perform your work in accordance with your training, written procedures, and supervisor directions.
- Immediately report any “near misses” or unsafe conditions to your faculty supervisor.
- Maintain a safe work environment for yourself and others.

COLLEGE AND DEPARTMENTS

- Provide supplementary online Safety training (college).
- Help supervisors maintain a safe working environment by providing necessary resources and support.
- Provide a college Occupational Health & Safety Specialist who serves as a resource those in the college can use for hazard identification, hazard evaluation, and mitigation and control selection. (College)
- Collect and maintain training records (Departments).
- Address issues identified by Risk Management (College and Departments).

12.0 APPENDICES

Appendix A HAZARD CHECKLIST

This hazard checklist is provided to help individuals identify hazards when performing a Job Hazard Analysis (JHA). A checklist like this is not comprehensive...rather it is a starting point. After completing the checklist more thought is needed to make sure there aren't other hazards. Ask questions such as the following when considering what hazards may be present:

- What are normal operating conditions?
- What abnormal operating conditions can occur and how? *For instance, what if someone sets the wrong temperature or pressure?*
- What mistakes can people make? *For example, what would happen if someone added too much of a particular reactant to a reaction?*
- Is contamination a problem and can it create a hazard?
- In what way could valves, gages, and other components fail and what would be the result of such failure?

HAZARDS

Safety Hazards

- ☐ Laceration/cut
- ☐ Amputation
- ☐ Crushing injury
- ☐ Impact injury
- ☐ Fall
- ☐ Arc blast
- ☐ Shock
- ☐ Electrocution
- ☐ Pressure above 1 atm
- ☐ Pressure below 1 atm
- ☐ Burn / Frostbite
- ☐ Explosion
- ☐ Flammable Gas
- ☐ Flammable Aerosol
- ☐ Oxidizing Gas
- ☐ Gas Under Pressure
- ☐ Flammable Liquid
- ☐ Flammable Solid
- ☐ Self-reacting Substance / Mixture
- ☐ Pyrophoric Liquid
- ☐ Pyrophoric Solid
- ☐ Self-heating Substance / Mixture
- ☐ Emits Flammable Gas When in Contact With Water
- ☐ Oxidizing Liquid
- ☐ Oxidizing Solid
- ☐ Organic Peroxide
- ☐ Corrosive to Metals
- ☐ Pressure (High / Low)
- ☐ Temperature (Hot / Cold)
- ☐ Hazardous Chemical Reaction
- ☐ Other:

Health Hazards

- ☐ Acutely Toxic
- ☐ Specific Target Organ Toxicity
- ☐ Toxic to Reproductive System
- ☐ Skin Corrosion / Irritation
- ☐ Serious Eye Damage / Irritation
- ☐ Respiratory Sensitizer
- ☐ Skin Sensitizer
- ☐ Carcinogenic
- ☐ Germ Cell Mutagenicity
- ☐ Aspiration Hazard
- ☐ Effects On or via Lactation
- ☐ Heat stress or shock
- ☐ Hypothermia
- ☐ Noise
- ☐ Ultraviolet radiation
- ☐ Infrared radiation
- ☐ Ionizing Radiation
- ☐ Excessive reaching
- ☐ Repetitive motion
- ☐ Cumulative trauma

Potential Exposure Routes

- ☐ Inhalation
- ☐ Skin Absorption
- ☐ Ingestion
- ☐ Injection

Appendix B

Template for Standard Operating Procedure (SOP)

Brigham Young University

SOP

Standard Operating Procedure

[Title]

Principal Investigator:

Primary Contact:

Secondary Contact:

Lab #:

Building:

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Important Phone Numbers:

University Police: 801-422-2222

Risk Management: 801-422-4468

Chemicals Management: 801-422-6382

College Health & Safety Officer: 801-422-6589

Poison Control: 800-222-1222

Completed By:

Edited By:

Approved By:

Description of Work and Scope of SOP

Prequalifications

Chemicals Used:

Summary of Physical Hazards

Summary of Chemical Hazards

Equipment Required to Perform Work

Engineering Controls

Personal Protective Equipment (PPE)

Step-by-Step Procedures for How to Safety Perform Work

A.

“Unwanted Lab Material” Collection Procedures

Summary of What Training is Required

Chemical Spill / Release Procedures

Attachment: Lab Layout

APPENDIX C

HAZARD EVALUATION TECHNIQUES

Qualitative Techniques: These methods help a multi-disciplinary team (1) identify potential accident scenarios and (2) evaluate the scenario in sufficient detail to make a reasonable judgment of risk. If the team is confused on the risk, a scenario identified in a qualitative hazard review may be further analyzed using one or more of the quantitative techniques.

Preliminary Hazard Analysis (PreHA): Often used to evaluate hazards early in the life of a process and is generally applied during the conceptual design or R&D phase. This method can be very useful when making site selection decisions. It is also commonly used as a design review tool before a process P&ID is developed.

Hazard and Operability (HAZOP) Analysis: Systematic identification of potential operating problems using a series of guide words to investigate deviations. Can be applied to any mode of operation of a flow process and can also be applied to any procedure or flowchart.

What-If Analysis: A brainstorming approach in which a group of experienced people familiar with the work process ask questions or voice concerns about possible undesired events. The method does not use guide words.

2 Guide Word Analysis: A systematic method in which potential operating problems are identified by asking what would happen if a step in a procedure were (1) skipped or (2) performed incorrectly. This method is applicable to any procedure (startup, shutdown, online maintenance, or normal batch operations), but does not apply to continuous operating mode.

Failure Modes and Effects Analysis (FMEA): A systematic, tabular method for evaluating and documenting the effects of known types of component failures. Applies to electrical/mechanical systems. Can also be applied to flow systems where very high reliability factors are needed (such as fire-fighting water supply systems).

Checklist (traditional): A detailed list of desired system attributes or steps for a system or operator to perform. Usually written from experience and used to assess the acceptability or status of the system or operation compared to established norms.

Quantitative Techniques: These do not identify possible accident scenarios, but they instead aid in risk judgment by provide more detailed, statistical evaluations of the risk of a specific scenario.

Layer of Protection Analysis (LOPA): A method that uses pre-defined values for initiating events, independent protection layers, and consequences to provide an order-of-magnitude estimate of risk. LOPA applies to a single cause-consequence pair. Scenarios are identified elsewhere (typically in a qualitative hazard evaluation).

Fault Tree Analysis (FTA): A logic model that graphically portrays the combinations of failures that can lead to a specific main failure or incident of interest (Top event). This method using

Boolean Logic (And & Or logic gates). Assigning statistical values to each end point on a branch allows the calculation of risk.

Event Tree Analysis (ETA): A logic model that graphically portrays the combinations of events and circumstances in an incident sequence. Assigning statistical values to each branch point (failure or condition) allows the calculation of composite risk starting from a defined initiating event.

Human Reliability Analysis (HRA) event tree: A graphical model of sequential events in which the tree limbs designate human actions and other events as well as different conditions or influences upon these events. Assigning statistical values to each branch point (correct or incorrect performance of a step) allows the calculation of composite risk starting from a defined first step.